

ARJAN WARAICH

647-473-1175 | waraicharjan97@gmail.com | linkedin.com/in/arjwaraich | github.com/RJN25 | devpost.com/RJN25

7+ Major League Hacking Hackathon Wins | \$20K+ in Prize Winnings | 1st Place: BramHacks, HackThe6ix, TerraHacks

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, Rust, SQL, R, MATLAB, Micropython, Verilog, HDL
ML/AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, LSTM, TabPFN, SHAP, GNNs, Transformers, YOLOv8, RT-DETR
Robotics & Embedded: ROS 2, Isaac Sim, Gazebo, FoxGlove, AMCL, SLAM, PID, **J1939/OBD-II CAN**, Arduino, OpenMV
Cloud & Tools: AWS (S3, EC2), Docker, D-Wave Ocean SDK, Flask, HuggingFace, Node.js, React, MongoDB, Git, Linux, FAISS
Vector DB

EXPERIENCE

- AI/ML Infrastructure Developer** Jun 2026 – Present
margIQ by OMRAM Fleet Solutions — AI-native transportation intelligence startup Remote
- Architected and coded **8 core WAYZ navigation engines**, modeling the North American road network as a weighted spatial graph to jointly optimize ETA, fuel cost, tolls/FX, dock detention, ferry crossings, and truck legality (height/length/axle/hazmat); paired **Dijkstra/A*** with **spatio-temporal GNNs**, **LightGBM** ETA regression, and **latent attention-based** behavioural routing; researching **D-Wave** quantum annealing for **VRP**
 - On the research engineering team, co-designed a custom edge gateway aggregating **CAN bus (J1939)/OBD-II** telemetry with AI dashcam streams, prioritizing **AWS S3 + EC2** for cloud inference, and helped develop **real-time encryption** for live vehicle tracking
 - Built **agentic** dashcam AI powering driver monitoring and coaching dashboards, combining **MediaPipe** fatigue/distraction tracking, **RT-DETR** object detection, and **temporal knowledge graphs** as long-term agent memory
- Robotics Simulation & ML Engineer** Sept. 2026 – Present
WATonomous | UWaterloo Humanoid Robotics Team Waterloo, ON
- Trained **Action Chunking Transformer (ACT)** visuomotor policies in **NVIDIA Isaac Sim**, using **ROS 2**, **PyTorch**, Python, and **C++** to map camera and robot-state observations to manipulation trajectories
 - Integrated learned policies into the ROS 2 robotics stack, developing perception, state estimation, **PID control**, trajectory interfaces, and **sim-to-real** evaluation pipelines for closed-loop deployment

CONFERENCE & RESEARCH PAPERS

- UWaterloo Machine Learning Researcher** Sept. 2026 – Present
Collaborative Research Team; preparing submission for AAAI 2027 Waterloo, ON
- Studied whether **explanation drift** (shifts in SHAP feature attributions) and embedding drift can act as **label-free early-warning signals** of model degradation under **distribution shift**
 - Benchmarked label-free monitoring signals (uncertainty, **SHAP** explanation drift, embedding drift, raw-input drift) across **7 tabular tasks in 5 domains** with **Logistic Regression**, **XGBoost**, and **TabPFN**; showed a **0.921 error-detection AUROC** score still alarmed **8 shift levels after a 10% macro-F1 drop**
 - Evaluated early warning over **800+ shift trajectories** with bootstrap CIs and **Wilcoxon (BH/Holm)** tests; uncertainty beat drift for all **3 model families**, while drift alarms hit **96–100% false-alarm rates**
- ProphetJet — Predictive Maintenance for NASA Turbofan Engine RUL** | *LSTM, XGBoost, Random Forest* **CUCAI 2025**
- Built a hybrid **LSTM–XGBoost–Random Forest** pipeline forecasting Remaining Useful Life (**RUL**) of turbofan engines on NASA's **C-MAPSS** dataset, achieving **87.4% accuracy** ($\pm 2\%$ std)
 - Authored and presented a paper accepted into the **CUCAI 2025** proceedings as a high-school student

PROJECTS

- Heimdall — Satellite-Repair Robotics** | *Gazebo, Deep Learning, Robotic Control* **BramHacks 2025: 1st/800+, \$5K**
- Trained **deep-learning** models forecasting satellite battery **SOH and RUL** degradation, plus **classifiers** identifying specific satellites for targeted repair planning
 - Simulated autonomous orbital docking in **Gazebo** with a learned **thruster-matrix control policy**, driving a **3D-printed** repair arm mounted on a satellite; presented to Brampton City Council
- Predictra — Industrial Machine-Health AI** | *LSTM Autoencoders, XGBoost, TensorFlow, React* **SpurHacks: 2nd/2,200+**
- Built an **unsupervised RUL** pipeline (**LSTM Autoencoders**, Bayesian bootstrapping, **XGBoost**) needing no labeled failure data, generalizing across **15+ machine types**; won the **\$5.5K Top AI-Startup Prize**
 - Engineered **anomaly detection** scoring per-timestep risk **[0–1]** via reconstruction-error thresholds, served over **JWT**-authenticated endpoints to a **React** dashboard
- TurretGuard — Drowning-Detection Robot** | *YOLOv8, Edge Impulse, OpenMV, Arduino* **HackThe6ix: 1st/600+**
- Trained **YOLOv8 (1,615+ images)** and **Edge Impulse FOMO (MobileNetV2)** models for real-time detection on an **OpenMV H7** at **<200ms**, driving an **Arduino** servo rescue turret over serial
- VEX Robotics 19109A** | *AMCL, SLAM, PID Control* **2× World Championship Qualifier**
- Developed **Adaptive Monte Carlo Localization (AMCL)**, **SLAM**, and **PID** controllers for autonomous navigation on a **world-stage** competition robot

EDUCATION

University of Waterloo — BAsC, Computer Engineering Sept. 2026 – Present
University of Toronto Schools — GPA: **4.0/4.0** (Math, CS, Physics, Chemistry) Sept. 2024 – June 2026